Deep Learning Projects

A collection of deep learning projects built from coursework, progressing from first-principles neural network fundamentals through to applied model building. Each project includes full derivations/implementation and, where relevant, numerical or empirical verification of correctness.

Projects

01 — Backpropagation from Scratch

A from-scratch implementation of reverse-mode automatic differentiation for a feedforward neural network, built with pure NumPy (no autograd libraries). Derives and implements every layer's backward pass from first principles, verified against finite-difference gradients to a relative error under 2e-5.

Skills: Manual backpropagation, Jacobian/vector-Jacobian products, gradient verification, NumPy

More projects are being added.

Why build from scratch?

Frameworks like PyTorch and TensorFlow abstract away the mechanics of gradient computation. Building backpropagation manually, deriving each gradient via perturbation analysis rather than using autograd, demonstrates a working understanding of what's actually happening during training, not just how to call `.backward()`.